

Unity requires physically enforced agreement

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Abstract

Human conscious experience is unified: the brain's many local descriptions of a shared situation agree. We argue that unity requires this agreement to be physically enforced, produced and maintained by a coupling through which the graded physical state of each region continuously pulls the regions it overlaps toward agreement. Clocked digital hardware computes agreement instead. Its noise margins make every content readout locally constant in micro state, and we prove that a readout with a margin, under continuous micro dynamics, depends on no region's graded state; content read from bits is therefore never physically enforced, whatever algorithm the bits run. For a coupled sheet of oscillators we show that the width of the coupling kernel sets how fast agreement decays with distance, and that only a kernel tail slower than r^{-4} makes it independent of distance. The premise yields interventional tests on cortex and on artificial systems, and we state the outcomes that would refute it on each.

1 Introduction

Premise P (physical unity). A system’s contents are unified only if the content of each region depends continuously on the graded physical state of the regions it overlaps, through a coupling that contracts their disagreement.

P is a necessary condition for unity. It rejects functionalism about unity: a perfect simulation of a coupled field reproduces the field’s content-level dynamics and fails P. The paper makes three claims:

1. A content readout with a noise margin, under continuous micro dynamics, depends on the graded state of no region, so content read from bits is not physically enforced whatever algorithm the bits run (Section 4).
2. On a noisy coupled sheet, the width of the coupling kernel sets how fast agreement decays with distance, and only a kernel tail slower than r^{-4} makes agreement independent of distance (Section 5).
3. P predicts that a sub-threshold perturbation of one cortical region shifts the content decoded in an overlapping region toward agreement, and that no perturbation below a digital system’s noise margin shifts any of its contents (Section 8).

2 Synchrony is not unity

Equal phases coexist with incompatible local descriptions, and compatible local descriptions determine a unique global state. Unity is therefore a property of what overlapping regions describe, and synchrony is at most a route to it.

3 Unity as agreement over a declared cover

Agreement is measured between regions that decode a shared variable. The cover of regions is fixed by decodability before agreement is measured, so that the measurement cannot be emptied by the choice of cover.

4 Physical enforcement versus routed agreement

A system is a micro dynamics f on the joint micro state $x = (x_i)_i$ of its regions, read by a content map π that assigns a content to each region. Its content level is closed when the next content is a function of the current one, $\pi \circ f = g \circ \pi$ for some content dynamics g . A readout has a noise margin on an operating set when every micro state in that set has a neighbourhood on which the readout is constant. Content depends gradedly on a region when no neighbourhood of that region's micro state leaves the next content unchanged, the others held fixed. The overlap discrepancy, a measure of disagreement between the contents of overlapping regions, contracts near a state when one step of f shrinks it by a factor below one throughout a neighbourhood of that state, so that perturbations below every threshold are pulled back toward agreement along with the rest. Agreement is physically enforced at a state when content depends gradedly on each region there and the overlap discrepancy contracts near it; P requires it at the states the system occupies while its contents are unified.

If the micro dynamics is continuous and carries a state into a margin's operating set, no region's micro state enters the next content gradedly (`not_physicallyEnforced_of_margin`). A thresholded continuous quantity has a margin off its threshold, and a finite word of such readouts has one too, so the conclusion holds for any content read from bits (`not_physicallyEnforced_of_bits`).

The theorem places the criterion one level below the content dynamics,

and the placement is forced. Contraction of disagreement at the content level is computable: a digital system running a consensus algorithm, or integrating a coupled field, has a content map whose successor contracts the disagreement between overlapping regions exactly as the field does. A criterion stated on the content dynamics would classify that system with the field. P is stated on the dependence of content on graded micro state, and there the two differ: the field's content moves with every graded change in the regions it overlaps, while the simulation's content is read from bits held inside their margins and moves with none of them.

This is the simulation objection, and P answers it by denial. A perfect simulation of a coupled cortical field reproduces the field's content-level trajectory and is not unified in the sense P requires, because the agreement in it is computed from contents rather than enforced by the graded state that carries them. That is the commitment P makes about unity, stated on the same footing as a workspace theory's identification of consciousness with broadcast: a premise whose consequences can be tested, which Section 8 does.

5 How far a physical coupling enforces agreement

At identical natural frequencies a noisy coupled sheet is a gradient system. In the harmonic approximation the variance of the phase difference between two sites is the noise strength times their effective resistance, with the coupling constants as conductances. Under short-range coupling that resistance grows as the logarithm of distance, with a coefficient inversely proportional to the kernel's second moment; a kernel tail slower than r^{-4} bounds it.

The logarithmic growth is a lower bound for every kernel of bounded range. Group the sites into shells around a site a , so that the coupling joins only sites in the same or adjacent shells. The shells then act as conduc-

tors in series, and the effective conductance from a to any site beyond shell n is at most $(\sum_{k < n} 1/C_k)^{-1}$, where C_k is the coupling crossing from shell k to shell $k + 1$ (`conductance_le_shell`). On a planar sheet the perimeter of shell k grows linearly in k , so $C_k \leq c(k + 1)$ for a constant c set by the kernel's strength and width, and the effective resistance is at least $\log(n + 1)/c$ (`conductance_mul_log_le`). Widening the kernel enlarges its shells and raises c , which slows the growth and leaves it unbounded. Distance-independent agreement on a sheet therefore needs coupling of unbounded range.

Simulated sheets at identical frequencies, of side 64 and 128, with nearest-neighbour, exponential and power-law kernels of tail exponent $\sigma = 1$ and $\sigma = 3$ at a common row sum, follow the harmonic identity in the ordered phase. The measured phase variance is at most 1.15 times $D R_{\text{eff}}$ at $D = 0.1$ and at most 1.26 times at $D = 0.4$, the excess rising with noise as the cosine coupling softens, and a logarithm fits its growth with distance better than a line in 16 of the 16 runs at those noise levels. At $D = 1.2$ the nearest-neighbour sheet disorders: the mean chord discrepancy $\langle 1 - \cos(\theta_a - \theta_b) \rangle$ at the largest separation is 1.00. The exponential and $\sigma = 1$ sheets stay ordered at that noise (0.46 and 0.50), and the $\sigma = 3$ sheet nearly disorders (0.92), so the kernel's width sets the order-disorder crossover as well as the rate of decay. Exact resistance sums at sheet sides 128, 512 and 1024 separate width from tail. The exponential kernel, with 25 times the nearest-neighbour stiffness, gives resistance at half the side of 0.531, 0.549 and 0.558, still growing; the $\sigma = 1$ tail gives 0.580, 0.577 and 0.577. Over a finite cortical extent a wide short-range kernel and a slow tail therefore differ by a slowly growing logarithm, and the measurement that separates them needs the kernel's second moment as well as its tail exponent.

Three facts hold without the harmonic approximation. Adding coupling never raises the effective resistance (`conductance_mono`), so a long-range tail added to a short-range kernel can only improve agreement, whatever its

shape. A direct coupling between two sites caps their resistance at its reciprocal, at any distance (`le_conductance_of_edge`). A Lipschitz encoder carries phase agreement to content agreement: the mean-square content discrepancy is at most the squared Lipschitz constant times the mean-square phase difference (`integral_dist_sq_le`). With the harmonic identity, an encoder of Lipschitz constant L at noise strength D therefore has root-mean-square content discrepancy at most $L\sqrt{D R_{\text{eff}}}$ between two sites of effective resistance R_{eff} .

Content can agree where phases differ. An encoder that reads the phases of a stencil around a site relative to the phase at that site reports the same content at every site of a plane wave, whatever its gradient, and at every site of a winding on a ring or torus (`relContent_eq_of_wave`). For a wave plus a departure, the content discrepancy between two sites of a Lipschitz relative encoder is at most its Lipschitz constant times the departure's relative readings at the two sites, with the gradient of the wave and the separation of the sites absent from the bound (`dist_relContent_le`). Travelling waves and windings, the configurations that attain the worst case of a hop-by-hop chain bound, thus cost a relative encoder nothing; what it pays for is the local roughness of the field.

6 The unity window

Agreement cannot arrive before the causal cone of the coupling allows, and disagreement decays at the relaxation rate of the coupling. Unity at a given time scale requires the interval between the two to be nonempty.

The lower edge is causal. A report at one site that tracks a variable at another to within ε , for two values of the variable more than 2ε apart, is possible after T rounds of the coupling only if the second site lies in the first site's causal past for T rounds (`mem_ball_of_reconstructs`). The upper edge is relaxation. If the coupling contracts the overlap discrepancy by a factor $\kappa < 1$

at every step on a set of states it keeps, a discrepancy starting at most d_0 is at most ε from step $\log(d_0/\varepsilon)/\log(1/\kappa)$ on (`disc_iterate_le_of_log_le`). The estimate is sharp in form: a discrepancy that shrinks by at most a factor ρ per step is still above ε at every earlier step than $\log(d_0/\varepsilon)/\log(1/\rho)$ (`lt_disc_iterate_of_lt_log`). The window opens at the later of the causal deadline and the relaxation time, and unity at the time scale of a content requires it to open before that content changes.

7 Current artificial systems

A transformer running on clocked digital hardware reads every content from bits held within their noise margins, so the result of Section 4 applies to it. Hardware whose contents are graded physical quantities coupled without barriers, such as analog, in-memory or neuromorphic substrates, can satisfy P.

The criterion concerns how a substrate is organised and says nothing of what it is made of. An analog crossbar whose column currents carry content and settle through shared conductances depends gradedly on every input voltage, and an array of coupled electronic or optical oscillators contracts phase disagreement through a physical coupling exactly as a cortical sheet does. A neuromorphic chip passes where its contents are read from membrane potentials coupled continuously, and fails where they are read from spike counts latched at a clock edge. Biological tissue passes or fails on the same terms: a neural circuit whose contents were read only from all-or-none events latched far from threshold would fall under the margin theorem as any digital circuit does.

Real digital hardware leaks: supply noise, crosstalk and temperature move the analog quantities behind every bit. The margin theorem locates where such leakage can reach content. If a content read as a word of thresholded continuous quantities depends gradedly on any region, some bit of the

next state sits exactly on its threshold (`exists_threshold_of_gradedDependence`), and correct operation keeps every state off that set. The margin also has a width. If every bit of the next state is at least δ from its threshold, each bit is read from an L_v -Lipschitz quantity and the micro dynamics is L -Lipschitz, a change to one region smaller than $\delta/(LL_v)$ changes no bit (`bits_eq_of_lipschitz`). Leakage below that scale carries no content; leakage above it flips a bit, which is a logical error. The nondeterminism of floating-point reduction order is a choice among logical trajectories and is outside both theorems: it varies which content is computed, not how content depends on graded state.

8 Tests

Graded cross-region dependence in cortex. Weak stimulation of one region below the behavioural threshold should shift the content decoded in an overlapping region, gradedly with stimulation strength and toward agreement with the stimulated region. Transcranial alternating current and weak intracortical fields supply the perturbation, and the decoded content is read from population activity in the overlapping region. An unchanged decoded content in the overlapping region, at stimulation strengths that measurably move the stimulated one, counts against P, since P then denies unity to the system whose unity motivates it.

Agreement as an attractor on both substrates. Perturb one region and measure whether agreement with its overlap partners is restored, and at what rate. In cortex the perturbation is stimulation and the measure is the disagreement between contents decoded from the perturbed region and its partners; P predicts restoration at the relaxation rate of Section 6, with a restoring pull proportional to the size of the perturbation below as well as above any threshold. In an artificial system the perturbation is activation

patching of one layer, head or position, applied at two scales: below the scale at which the logical trajectory changes, and above it. A digital system can restore agreement above that scale, by computation; below it the margin theorem predicts that nothing moves at all. The outcome that counts against the thesis on the artificial side is a graded shift of decoded content in one region, toward agreement, driven by an analog perturbation of another that stays inside every bit's margin, such as a supply or timing perturbation below the error threshold.

The field's causal cone. A quasi-static extracellular field reaches every point of the tissue it spans within one field update, while synaptic influence between distant regions arrives only after axonal conduction and synaptic transmission. Re-agreement between distant regions after a local perturbation, faster than the synaptic route allows, would identify a non-synaptic contribution to the coupling and measure the reach of the kernel tail of Section 5 directly.

Separating coupling from information. In every manipulation of cortex available so far, graded coupling and information flow move together: anaesthesia and callosal section remove both. A test that confirms P against functionalism must move one with the other held fixed. The candidate design applies weak alternating stimulation that shifts the coherence between two regions while a measured information-transfer quantity between them, such as transfer entropy estimated from their recorded activity, is held matched across conditions, and asks whether the agreement between the contents decoded from the two regions follows the coupling or the information. Agreement that follows coupling at matched information supports P over functionalism; agreement that follows information at matched coupling counts against P.

What counts as evidence about the phenomenal field. Every test above measures decoded agreement, which is agreement of the contents available to readout. If disunity of readout could always be set aside as disunity of access alone, no observation would bear on P. The observations that count as evidence about the phenomenal field are therefore fixed in advance: responses that integrate stimuli across the regions in question when the task requires it, and reports of a single field of experience spanning them. A dissociation in which such responses and reports split while graded coupling between the regions is intact, or stay integrated after graded coupling between them is removed, is accepted as evidence against P.

9 Limitations

P is a premise and is not derived from the dynamics; the formal results show what follows from it and not that it holds. It is necessary for unity and not sufficient for consciousness: coupled pendulums satisfy it. The margin theorem concerns idealised digital readouts, and graded leakage in real hardware that carries content-relevant agreement would narrow its reach. The identification of phase variance with effective resistance rests on the harmonic approximation at identical natural frequencies. The logarithmic lower bound on resistance is proved for kernels of bounded range; that it is attained, and that a tail slower than r^{-4} bounds the resistance, are established numerically on finite sheets. The unity window’s edges are proved for a given contraction factor; the identification of that factor with the coupling’s spectral gap is linear response, and whether the window is nonempty at cortical time scales awaits a magnitude estimate. The step from phase agreement to content agreement assumes one oscillator population per content patch, which is a modelling choice about cortex. A long-range tail improves agreement only in proportion to its conductance against the short-range part, and the measured endogenous fields of cortex are weak, so the tail’s contribution

at cortical scale is a quantity to be measured. Above the order-disorder crossover of Section 5 the harmonic identity fails and agreement is lost at every distance. Human data can falsify P but cannot by themselves confirm it against functionalism, because graded coupling and information flow co-vary in every manipulation available so far.